Linux Engineer Command Reference

A concise practical reference covering the most important Linux commands for software, AI/ML, DevOps, cloud, and backend engineers.

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1. Navigation

Command	Description
`pwd`	Show current directory
`ls`	List files and directories
`ls -l`	Detailed listing
`ls -la`	Detailed listing including hidden files
`cd DIR`	Change directory
`cd ..`	Go to parent directory
`cd .`	Stay in current directory
`cd ~`	Go to home directory
`cd /`	Go to filesystem root
`cd -`	Go to previous directory

```
pwd
ls -la
cd /home/sheharyar
cd ..
cd ~
```

---

2. Linux Paths

```
/     → filesystem root
~     → current user's home
.     → current directory
..    → parent directory
```

Absolute Path

/home/sheharyar/project/main.py

Starts from `/`.

Relative Path

project/main.py

Starts from the current directory.

---

3. Files and Directories

Command	Description
`mkdir DIR`	Create directory
`mkdir -p a/b/c`	Create nested directories
`touch FILE`	Create empty file
`cp FILE DEST`	Copy file
`cp -r DIR DEST`	Copy directory
`mv OLD NEW`	Rename/move
`rm FILE`	Delete file
`rm -r DIR`	Delete directory recursively
`rmdir DIR`	Delete empty directory
`rm -i FILE`	Ask before deleting
`rm -rf DIR`	Force recursive deletion

Examples:

mkdir project
touch main.py
cp main.py backup.py
mv backup.py backup_main.py
rm backup_main.py

---

4. Viewing Files

Command	Description
`cat FILE`	Display entire file
`less FILE`	Read file page by page
`head FILE`	Show beginning
`head -n 20 FILE`	Show first 20 lines
`tail FILE`	Show end
`tail -n 20 FILE`	Show last 20 lines
`tail -f FILE`	Follow a changing file
`nl FILE`	Show line numbers

Examples:

cat README.md
less README.md
head -n 10 app.log
tail -f app.log

Press `q` to exit `less`.

---

5. Searching Files and Text

`find`

Search for files/directories.

find . -name "main.py"

Search by type:

find . -type f
find . -type d

Search by extension:

find . -name "*.py"

`grep`

Search text inside files.

grep "error" app.log

Useful options:

grep -i "error" app.log
grep -r "TODO" .
grep -n "error" app.log

-i → ignore case
-r → recursive
-n → show line number

---

6. File Information

Command	Description
`file FILE`	Identify file type

`stat FILE`	Detailed file metadata
`wc FILE`	Count lines/words/bytes
`du -h`	Disk usage
`df -h`	Filesystem disk space

Examples:

file model.pkl
stat model.pkl
du -sh project/
df -h

---

7. Permissions

Linux permissions are commonly represented as:

r → read
w → write
x → execute

Example:

-rwxr-xr--

Change permissions:

chmod 755 script.sh
chmod +x script.sh
chmod -x script.sh

Common permissions:

755 → rwxr-xr-x
644 → rw-r--r--
600 → rw-------

---

8. Ownership

Check ownership:

ls -l

Change owner:

sudo chown user file.txt

Change owner and group:

sudo chown user:group file.txt

Change directory recursively:

sudo chown -R user:group project/

Change group:

sudo chgrp group file.txt

---

9. Users

Command	Description
`whoami`	Current username
`id`	UID, GID and groups
`groups`	Current user's groups
`who`	Logged-in users
`w`	Logged-in users + activity
`adduser USER`	Create user
`deluser USER`	Delete user

`passwd USER`	Change password
`usermod`	Modify user

Examples:

whoami
id
groups
sudo adduser ali
sudo passwd ali

---

10. Root and Sudo

`root` is the Linux superuser.

sudo command

Run a command with elevated privileges.

sudo apt update
sudo systemctl restart ssh
sudo whoami

Become root temporarily:

sudo -i

Leave root:

exit

Switch user:

su - username

---

11. Processes

A process is a running program.

Command	Description
`ps`	Show processes
`ps aux`	Detailed process list
`top`	Real-time process monitor
`htop`	Interactive process monitor
`pgrep NAME`	Find process
`kill PID`	Terminate process
`kill -9 PID`	Force terminate
`pkill NAME`	Kill by name

Examples:

ps aux
pgrep python
kill 1234

Use `kill -9` only when normal termination doesn't work.

---

12. Background and Foreground Jobs

Run a command in background:

python app.py &

Show jobs:

jobs

Bring job to foreground:

fg

Suspend a running program:

Ctrl + Z

Resume in background:

bg

---

13. Environment Variables

View variables:

env
printenv

Show one variable:

echo $PATH

Create variable:

export API_KEY="value"

Remove variable:

unset API_KEY

Common variables:

PATH
HOME
USER
SHELL
PWD

---

14. PATH

Check PATH:

echo $PATH

Temporarily add directory:

export PATH="$PATH:/my/program"

Linux searches directories in `$PATH` when you run commands.

---

15. Command Location

Find where a command comes from:

which python

Better diagnostic:

type python

Find executable:

whereis python

---

16. Command History

history

Search history:

Ctrl + R

Run previous command:

!!

Run command number:

!123

Clear history:

history -c

---

17. Terminal Management

Command	Description
`clear`	Clear terminal
`reset`	Reset terminal
`echo`	Print text
`printf`	Formatted output
`date`	Current date/time
`whoami`	Current user
`hostname`	Computer hostname

---

18. Package Management — Ubuntu

Ubuntu uses `apt`.

Update package information:

sudo apt update

Upgrade packages:

sudo apt upgrade

Install:

sudo apt install PACKAGE

Remove:

sudo apt remove PACKAGE

Search:

apt search PACKAGE

Show package information:

apt show PACKAGE

Remove unused packages:

sudo apt autoremove

Examples:

sudo apt install tree
sudo apt install git
sudo apt install curl

---

19. Archives

tar

Create archive:

tar -cf archive.tar project/

Extract:

tar -xf archive.tar

Create gzip archive:

tar -czf project.tar.gz project/

Extract gzip:

tar -xzf project.tar.gz

List contents:

tar -tf archive.tar

---

20. Compression

gzip file.txt
gunzip file.txt.gz

For ZIP:

zip archive.zip file.txt
unzip archive.zip

Install ZIP tools if necessary:

sudo apt install zip unzip

---

21. Disk Management

Check filesystem usage:

df -h

Check directory size:

du -sh project/

Check individual items:

du -sh *

List block devices:

lsblk

---

22. System Information

uname -a

Kernel/system information.

hostnamectl

System and hostname information.

lscpu

CPU information.

free -h

RAM usage.

lsmem

Memory information.

lsblk

Storage devices.

---

23. Operating System Information

cat /etc/os-release

Shows Linux distribution information.

Example:

Ubuntu

Kernel version:

uname -r

---

24. Services — systemd

Check service:

systemctl status SERVICE

Start:

sudo systemctl start SERVICE

Stop:

sudo systemctl stop SERVICE

Restart:

sudo systemctl restart SERVICE

Enable at boot:

sudo systemctl enable SERVICE

Disable at boot:

sudo systemctl disable SERVICE

Examples:

sudo systemctl status ssh
sudo systemctl restart docker

---

25. Logs

System logs:

journalctl

Follow logs:

journalctl -f

Service logs:

journalctl -u SERVICE

Example:

journalctl -u ssh

Recent logs:

journalctl -n 50

---

26. Networking

Show IP addresses:

ip addr

Short version:

ip a

Show routes:

ip route

Test connectivity:

ping google.com

DNS lookup:

nslookup google.com

or:

dig google.com

Check open/listening ports:

ss -tulpn

---

27. curl

`curl` transfers data over protocols such as HTTP/HTTPS.

GET request:

curl https://example.com

Download file:

curl -O https://example.com/file.zip

Follow redirects:

curl -L URL

API request:

curl -X GET https://api.example.com/users

---

28. wget

Download files:

wget URL

Example:

wget https://example.com/file.zip

---

29. SSH

SSH provides secure remote shell access.

Connect:

ssh user@server

Example:

ssh ubuntu@192.168.1.10

Specify key:

ssh -i key.pem user@server

Copy file to server:

scp file.txt user@server:/home/user/

Copy from server:

scp user@server:/home/user/file.txt .

---

30. SSH Keys

Generate key:

ssh-keygen

Copy public key:

ssh-copy-id user@server

Typical files:

~/.ssh/id_ed25519
~/.ssh/id_ed25519.pub

Never share your private key.

---

31. Git

Check Git:

git --version

Initialize repository:

git init

Clone:

git clone URL

Check status:

git status

Stage:

git add .

Commit:

git commit -m "message"

Push:

git push

Pull:

git pull

Branches:

git branch
git switch -c feature
git switch main

---

32. Symbolic Links

Create symlink:

ln -s TARGET LINK

Example:

ln -s /var/www/project ~/project

View links:

ls -l

---

33. Redirection

Output to file:

command > file.txt

Append:

command >> file.txt

Input:

command < file.txt

Error redirection:

command 2> error.log

Output + error:

command > output.log 2>&1

---

34. Pipes

A pipe sends one command's output to another command.

command1 | command2

Example:

ps aux | grep python

Another example:

ls -la | less

---

35. Useful Text Tools

`sort`

sort file.txt

Sort lines.

`uniq`

sort file.txt | uniq

Remove adjacent duplicates.

cut

cut -d: -f1 /etc/passwd

Extract fields.

awk

awk '{print $1}' file.txt

Process structured text.

sed

sed 's/old/new/g' file.txt

Replace text.

---

36. xargs

Pass output as command arguments.

find . -name "*.log" | xargs rm

Use carefully when deleting files.

---

37. Shell Scripting

Create:

touch script.sh

Make executable:

chmod +x script.sh

Run:

./script.sh

Basic script:

#!/bin/bash

echo "Hello Linux"

Variables:

```
name="Sheharyar"
echo "$name"
```

Arguments:

```
echo "$1"
```

Condition:

```
if [ "$1" = "hello" ]; then
    echo "Hello"
fi
```

Loop:

```
for file in *.txt; do
    echo "$file"
done
```

---

38. Bash Operators

Logical AND:

command1 && command2

Run second only if first succeeds.

Logical OR:

command1 || command2

Run second if first fails.

Sequential:

command1 ; command2

---

39. File Descriptors

Linux commonly uses:

0 → stdin
1 → stdout
2 → stderr

Example:

command > output.txt

Redirect stdout.

command 2> error.txt

Redirect stderr.

command &> output.txt

Redirect stdout and stderr in Bash.

---

40. Process Signals

Common signals:

SIGTERM → 15 → graceful termination
SIGKILL → 9 → force termination
SIGINT → 2 → interrupt
SIGHUP → 1 → hangup

Examples:

kill PID
kill -9 PID

Prefer normal `kill`/SIGTERM before `kill -9`.

---

41. Jobs

List jobs:

jobs

Background:

command &

Suspend:

Ctrl + Z

Resume background:

bg

Foreground:

fg

---

42. Cron Jobs

Edit scheduled tasks:

crontab -e

View:

crontab -l

Cron example:

0 2 * * * /home/user/backup.sh

Runs the script every day at 2:00 AM.

---

43. Time and Scheduling

date

Show current date/time.

sleep 5

Wait 5 seconds.

---

44. Users and Permissions Files

Important files:

/etc/passwd → user accounts
/etc/shadow → password hashes
/etc/group → groups
/etc/sudoers → sudo configuration

Don't casually edit `/etc/sudoers`.

Use:

sudo visudo

when modifying sudo configuration.

---

45. Important System Directories

/ → filesystem root
/home → user home directories
/root → root's home
/etc → configuration
/var → variable data/logs
/tmp → temporary files
/usr → applications/libraries
/opt → optional software
/dev → devices
/proc → process/kernel information
/sys → kernel/device information
/mnt → temporary mounts
/media → removable media

---

46. Mounting

Show mounted filesystems:

mount

Show filesystem usage:

df -h

Mount a filesystem:

sudo mount DEVICE DIRECTORY

Unmount:

sudo umount DIRECTORY

---

47. Environment and Shell

Current shell:

echo $SHELL

Current home:

echo $HOME

Current user:

echo $USER

Current directory:

echo $PWD

Reload Bash configuration:

source ~/.bashrc

---

48. Aliases

Create temporary alias:

alias ll='ls -la'

Remove:

unalias ll

Permanent aliases can be placed in:

~/.bashrc

---

49. `man`

Linux manual pages:

man COMMAND

Examples:

man ls
man cp
man chmod

Search manuals:

man -k keyword

Exit:

q

---

50. Help

Command help:

command --help

Example:

cp --help

For Bash built-ins:

help cd

---

51. Terminal Shortcuts

Shortcut	Description
`Ctrl + C`	Stop current command
`Ctrl + Z`	Suspend process
`Ctrl + D`	Exit shell / EOF
`Ctrl + L`	Clear terminal
`Ctrl + R`	Search command history
`Ctrl + A`	Beginning of line
`Ctrl + E`	End of line
`Ctrl + U`	Delete before cursor
`Ctrl + K`	Delete after cursor
`Tab`	Autocomplete

---

52. Docker

Check:

docker --version

List containers:

docker ps
docker ps -a

List images:

docker images

Pull image:

docker pull IMAGE

Run container:

docker run IMAGE

Build:

docker build -t myapp .

Stop:

docker stop CONTAINER

Remove:

docker rm CONTAINER

View logs:

docker logs CONTAINER

Execute command inside container:

docker exec -it CONTAINER bash

---

53. Python Environment

Check Python:

python3 --version

Create virtual environment:

python3 -m venv .venv

Activate:

source .venv/bin/activate

Deactivate:

deactivate

Install:

pip install PACKAGE

Requirements:

pip install -r requirements.txt

---

54. Node.js

Check:

node --version
npm --version

Install packages:

npm install

Run project:

npm run dev

---

55. AI/ML Useful Commands

Check NVIDIA GPU:

nvidia-smi

Check CUDA compiler:

nvcc --version

Check GPU processes:

nvidia-smi

Check Python:

python3 --version

Check installed Python packages:

pip list

This is especially useful when working with:

PyTorch
TensorFlow
CUDA
NVIDIA GPUs
Docker
Python

---

56. Logs for AI/Production Debugging

Follow application logs:

tail -f app.log

Search errors:

grep -i "error" app.log

Search recursively:

grep -ri "error" .

Check system logs:

journalctl -xe

Check a service:

journalctl -u SERVICE

---

57. Network Debugging

Check IP:

ip a

Check routes:

ip route

Test host:

ping example.com

Test HTTP:

curl -I https://example.com

DNS:

dig example.com

Listening ports:

ss -tulpn

Test specific port:

nc -zv HOST PORT

---

58. Common Engineering Workflow

```
cd project/
ls -la
git status
git pull

source .venv/bin/activate

python3 app.py

git add .
git commit -m "update"
git push
```

For a server:

```
ssh user@server
cd /app
git pull
sudo systemctl restart app
sudo systemctl status app
journalctl -u app -f
```

---

59. Essential Commands to Memorize

Navigation

```
pwd
ls
cd
```

Files

```
mkdir
touch
cp
mv
rm
```

Viewing

cat
less
head
tail

Search

find
grep

Permissions

chmod
chown
sudo

Users

whoami
id
groups

Processes

ps
top
htop
kill

Packages

apt

Services

systemctl
journalctl

Networking

ip
ping
curl
ss

Remote servers

ssh
scp

Development

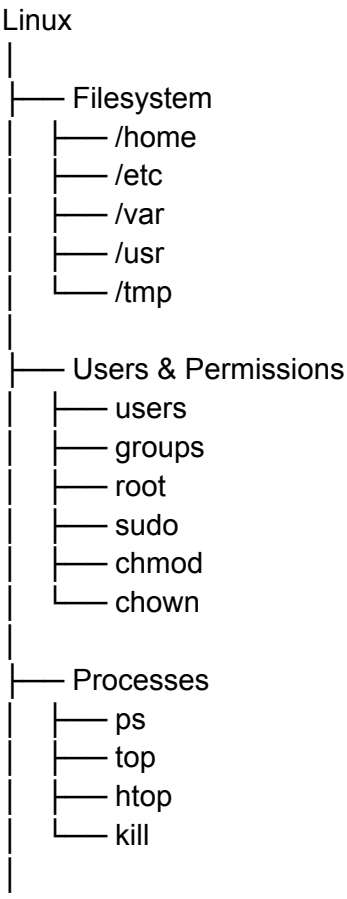
git
python3
pip
node
npm
docker

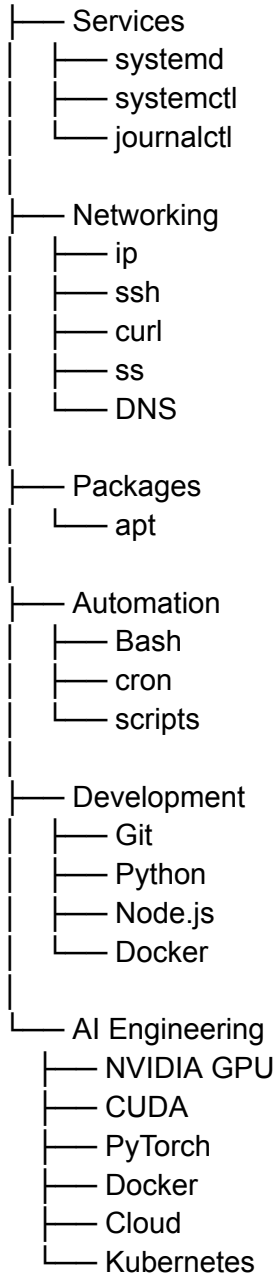
Help

man
--help

---

60. Linux Engineer Mental Model





Priority for an AI/Software Engineer

Learn in this order:

1. pwd / ls / cd
2. mkdir / touch / cp / mv / rm
3. cat / less / head / tail
4. find / grep
5. Absolute & relative paths
6. Users / groups / root
7. sudo
8. chmod / chown

9. Processes / ps / top / kill
10. Environment variables
11. apt
12. systemctl / journalctl
13. Networking / SSH
14. Bash scripting
15. Git
16. Docker
17. Cloud Linux servers
18. Kubernetes
19. NVIDIA / CUDA
20. AI/ML deployment

Core principle: Don't try to memorize every command. Learn the command families and use `man COMMAND` or `COMMAND --help` when you need the exact options.

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---

1. Navigation

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`cd -`	Go to previous directory

pwd
ls -la
cd /home/sheharyar
cd ..
cd ~

---

2. Linux Paths

/ → filesystem root
~ → current user's home
. → current directory
.. → parent directory

Absolute Path

/home/sheharyar/project/main.py

Starts from `/`.

Relative Path

project/main.py

Starts from the current directory.

---

3. Files and Directories

Command	Description
`mkdir DIR`	Create directory
`mkdir -p a/b/c`	Create nested directories
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`cp FILE DEST`	Copy file
`cp -r DIR DEST`	Copy directory
`mv OLD NEW`	Rename/move
`rm FILE`	Delete file
`rm -r DIR`	Delete directory recursively
`rmdir DIR`	Delete empty directory
`rm -i FILE`	Ask before deleting
`rm -rf DIR`	Force recursive deletion

Examples:

```
mkdir project
touch main.py
cp main.py backup.py
mv backup.py backup_main.py
rm backup_main.py
```

---

4. Viewing Files

Command	Description
`cat FILE`	Display entire file
`less FILE`	Read file page by page
`head FILE`	Show beginning
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`tail -n 20 FILE`	Show last 20 lines

`tail -f FILE`	Follow a changing file
`nl FILE`	Show line numbers

Examples:

cat README.md
less README.md
head -n 10 app.log
tail -f app.log

Press `q` to exit `less`.

---

5. Searching Files and Text

`find`

Search for files/directories.

find . -name "main.py"

Search by type:

find . -type f
find . -type d

Search by extension:

find . -name "*.py"

`grep`

Search text inside files.

grep "error" app.log

Useful options:

grep -i "error" app.log
grep -r "TODO" .
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-i → ignore case
-r → recursive
-n → show line number

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Examples:

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stat model.pkl
du -sh project/
df -h

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Linux permissions are commonly represented as:

r → read
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x → execute

Example:

-rwxr-xr--

Change permissions:

chmod 755 script.sh
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644 → rw-r--r--
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Check ownership:

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Change owner:

sudo chown user file.txt

Change owner and group:

sudo chown user:group file.txt

Change directory recursively:

sudo chown -R user:group project/

Change group:

sudo chgrp group file.txt

---

9. Users

Command	Description
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`id`	UID, GID and groups
`groups`	Current user's groups
`who`	Logged-in users
`w`	Logged-in users + activity
`adduser USER`	Create user
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`passwd USER`	Change password
`usermod`	Modify user

Examples:

whoami
id
groups
sudo adduser ali
sudo passwd ali

---

10. Root and Sudo

`root` is the Linux superuser.

sudo command

Run a command with elevated privileges.

sudo apt update
sudo systemctl restart ssh
sudo whoami

Become root temporarily:

sudo -i

Leave root:

exit

Switch user:

su - username

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11. Processes

A process is a running program.

Command	Description
`ps`	Show processes
`ps aux`	Detailed process list
`top`	Real-time process monitor
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`pgrep NAME`	Find process
`kill PID`	Terminate process
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Examples:

ps aux
pgrep python
kill 1234

Use `kill -9` only when normal termination doesn't work.

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12. Background and Foreground Jobs

Run a command in background:

python app.py &

Show jobs:

jobs

Bring job to foreground:

fg

Suspend a running program:

Ctrl + Z

Resume in background:

bg

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13. Environment Variables

View variables:

env
printenv

Show one variable:

echo $PATH

Create variable:

export API_KEY="value"

Remove variable:

unset API_KEY

Common variables:

PATH
HOME
USER
SHELL
PWD

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14. PATH

Check PATH:

echo $PATH

Temporarily add directory:

export PATH="$PATH:/my/program"

Linux searches directories in `$PATH` when you run commands.

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15. Command Location

Find where a command comes from:

which python

Better diagnostic:

type python

Find executable:

whereis python

---

16. Command History

history

Search history:

Ctrl + R

Run previous command:

!!

Run command number:

!123

Clear history:

history -c

---

17. Terminal Management

Command	Description
`clear`	Clear terminal
`reset`	Reset terminal
`echo`	Print text
`printf`	Formatted output
`date`	Current date/time
`whoami`	Current user

`hostname`	Computer hostname

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18. Package Management — Ubuntu

Ubuntu uses `apt`.

Update package information:

sudo apt update

Upgrade packages:

sudo apt upgrade

Install:

sudo apt install PACKAGE

Remove:

sudo apt remove PACKAGE

Search:

apt search PACKAGE

Show package information:

apt show PACKAGE

Remove unused packages:

sudo apt autoremove

Examples:

sudo apt install tree
sudo apt install git
sudo apt install curl

---

19. Archives

tar

Create archive:

tar -cf archive.tar project/

Extract:

tar -xf archive.tar

Create gzip archive:

tar -czf project.tar.gz project/

Extract gzip:

tar -xzf project.tar.gz

List contents:

tar -tf archive.tar

---

20. Compression

gzip file.txt
gunzip file.txt.gz

For ZIP:

zip archive.zip file.txt
unzip archive.zip

Install ZIP tools if necessary:

sudo apt install zip unzip

---

21. Disk Management

Check filesystem usage:

df -h

Check directory size:

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Check individual items:

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List block devices:

lsblk

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Kernel/system information.

hostnamectl

System and hostname information.

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CPU information.

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RAM usage.

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Memory information.

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Storage devices.

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cat /etc/os-release

Shows Linux distribution information.

Example:

Ubuntu

Kernel version:

uname -r

---

24. Services — systemd

Check service:

systemctl status SERVICE

Start:

sudo systemctl start SERVICE

Stop:

sudo systemctl stop SERVICE

Restart:

sudo systemctl restart SERVICE

Enable at boot:

sudo systemctl enable SERVICE

Disable at boot:

sudo systemctl disable SERVICE

Examples:

sudo systemctl status ssh
sudo systemctl restart docker

---

25. Logs

System logs:

journalctl

Follow logs:

journalctl -f

Service logs:

journalctl -u SERVICE

Example:

journalctl -u ssh

Recent logs:

journalctl -n 50

---

26. Networking

Show IP addresses:

ip addr

Short version:

ip a

Show routes:

ip route

Test connectivity:

ping google.com

DNS lookup:

nslookup google.com

or:

dig google.com

Check open/listening ports:

ss -tulpn

---

27. curl

`curl` transfers data over protocols such as HTTP/HTTPS.

GET request:

curl https://example.com

Download file:

curl -O https://example.com/file.zip

Follow redirects:

curl -L URL

API request:

curl -X GET https://api.example.com/users

---

28. wget

Download files:

wget URL

Example:

wget https://example.com/file.zip

---

29. SSH

SSH provides secure remote shell access.

Connect:

ssh user@server

Example:

ssh ubuntu@192.168.1.10

Specify key:

ssh -i key.pem user@server

Copy file to server:

scp file.txt user@server:/home/user/

Copy from server:

scp user@server:/home/user/file.txt .

---

30. SSH Keys

Generate key:

ssh-keygen

Copy public key:

ssh-copy-id user@server

Typical files:

~/.ssh/id_ed25519
~/.ssh/id_ed25519.pub

Never share your private key.

---

31. Git

Check Git:

git --version

Initialize repository:

git init

Clone:

git clone URL

Check status:

git status

Stage:

git add .

Commit:

git commit -m "message"

Push:

git push

Pull:

git pull

Branches:

git branch
git switch -c feature
git switch main

---

32. Symbolic Links

Create symlink:

ln -s TARGET LINK

Example:

ln -s /var/www/project ~/project

View links:

ls -l

---

33. Redirection

Output to file:

command > file.txt

Append:

command >> file.txt

Input:

command < file.txt

Error redirection:

command 2> error.log

Output + error:

command > output.log 2>&1

---

34. Pipes

A pipe sends one command's output to another command.

command1 | command2

Example:

ps aux | grep python

Another example:

ls -la | less

---

35. Useful Text Tools

`sort`

sort file.txt

Sort lines.

`uniq`

sort file.txt | uniq

Remove adjacent duplicates.

`cut`

cut -d: -f1 /etc/passwd

Extract fields.

`awk`

awk '{print $1}' file.txt

Process structured text.

`sed`

sed 's/old/new/g' file.txt

Replace text.

---

36. `xargs`

Pass output as command arguments.

find . -name "*.log" | xargs rm

Use carefully when deleting files.

---

37. Shell Scripting

Create:

touch script.sh

Make executable:

chmod +x script.sh

Run:

./script.sh

Basic script:

#!/bin/bash

echo "Hello Linux"

Variables:

```
name="Sheharyar"
echo "$name"
```

Arguments:

echo "$1"

Condition:

```
if [ "$1" = "hello" ]; then
    echo "Hello"
fi
```

Loop:

```
for file in *.txt; do
    echo "$file"
done
```

---

38. Bash Operators

Logical AND:

command1 && command2

Run second only if first succeeds.

Logical OR:

command1 || command2

Run second if first fails.

Sequential:

command1 ; command2

---

39. File Descriptors

Linux commonly uses:

0 → stdin
1 → stdout
2 → stderr

Example:

command > output.txt

Redirect stdout.

command 2> error.txt

Redirect stderr.

command &> output.txt

Redirect stdout and stderr in Bash.

---

40. Process Signals

Common signals:

SIGTERM → 15 → graceful termination
SIGKILL → 9 → force termination
SIGINT → 2 → interrupt
SIGHUP → 1 → hangup

Examples:

kill PID
kill -9 PID

Prefer normal `kill`/SIGTERM before `kill -9`.

---

41. Jobs

List jobs:

jobs

Background:

command &

Suspend:

Ctrl + Z

Resume background:

bg

Foreground:

fg

---

42. Cron Jobs

Edit scheduled tasks:

crontab -e

View:

crontab -l

Cron example:

0 2 * * * /home/user/backup.sh

Runs the script every day at 2:00 AM.

---

43. Time and Scheduling

date

Show current date/time.

sleep 5

Wait 5 seconds.

---

44. Users and Permissions Files

Important files:

/etc/passwd → user accounts
/etc/shadow → password hashes
/etc/group → groups
/etc/sudoers → sudo configuration

Don't casually edit `/etc/sudoers`.

Use:

sudo visudo

when modifying sudo configuration.

---

45. Important System Directories

/ → filesystem root
/home → user home directories
/root → root's home
/etc → configuration
/var → variable data/logs
/tmp → temporary files
/usr → applications/libraries
/opt → optional software
/dev → devices
/proc → process/kernel information
/sys → kernel/device information
/mnt → temporary mounts
/media → removable media

---

46. Mounting

Show mounted filesystems:

mount

Show filesystem usage:

df -h

Mount a filesystem:

sudo mount DEVICE DIRECTORY

Unmount:

sudo umount DIRECTORY

---

47. Environment and Shell

Current shell:

echo $SHELL

Current home:

echo $HOME

Current user:

echo $USER

Current directory:

echo $PWD

Reload Bash configuration:

source ~/.bashrc

---

48. Aliases

Create temporary alias:

alias ll='ls -la'

Remove:

unalias ll

Permanent aliases can be placed in:

~/.bashrc

---

49. `man`

Linux manual pages:

man COMMAND

Examples:

man ls
man cp
man chmod

Search manuals:

man -k keyword

Exit:

q

---

50. Help

Command help:

command --help

Example:

cp --help

For Bash built-ins:

help cd

---

51. Terminal Shortcuts

Shortcut	Description
`Ctrl + C`	Stop current command
`Ctrl + Z`	Suspend process
`Ctrl + D`	Exit shell / EOF
`Ctrl + L`	Clear terminal
`Ctrl + R`	Search command history
`Ctrl + A`	Beginning of line
`Ctrl + E`	End of line
`Ctrl + U`	Delete before cursor
`Ctrl + K`	Delete after cursor
`Tab`	Autocomplete

---

52. Docker

Check:

docker --version

List containers:

docker ps
docker ps -a

List images:

docker images

Pull image:

docker pull IMAGE

Run container:

docker run IMAGE

Build:

docker build -t myapp .

Stop:

docker stop CONTAINER

Remove:

docker rm CONTAINER

View logs:

docker logs CONTAINER

Execute command inside container:

docker exec -it CONTAINER bash

---

53. Python Environment

Check Python:

python3 --version

Create virtual environment:

python3 -m venv .venv

Activate:

source .venv/bin/activate

Deactivate:

deactivate

Install:

pip install PACKAGE

Requirements:

pip install -r requirements.txt

---

54. Node.js

Check:

node --version
npm --version

Install packages:

npm install

Run project:

npm run dev

---

55. AI/ML Useful Commands

Check NVIDIA GPU:

nvidia-smi

Check CUDA compiler:

nvcc --version

Check GPU processes:

nvidia-smi

Check Python:

python3 --version

Check installed Python packages:

pip list

This is especially useful when working with:

PyTorch
TensorFlow
CUDA
NVIDIA GPUs
Docker
Python

---

56. Logs for AI/Production Debugging

Follow application logs:

tail -f app.log

Search errors:

grep -i "error" app.log

Search recursively:

grep -ri "error" .

Check system logs:

journalctl -xe

Check a service:

journalctl -u SERVICE

---

57. Network Debugging

Check IP:

ip a

Check routes:

ip route

Test host:

ping example.com

Test HTTP:

curl -I https://example.com

DNS:

dig example.com

Listening ports:

ss -tulpn

Test specific port:

nc -zv HOST PORT

---

58. Common Engineering Workflow

cd project/
ls -la
git status
git pull

source .venv/bin/activate

python3 app.py

git add .
git commit -m "update"
git push

For a server:

ssh user@server
cd /app
git pull
sudo systemctl restart app
sudo systemctl status app
journalctl -u app -f

---

59. Essential Commands to Memorize

Navigation

pwd
ls
cd

Files

mkdir
touch
cp
mv
rm

Viewing

cat
less
head
tail

Search

find
grep

Permissions

chmod
chown
sudo

Users

whoami
id
groups

Processes

ps
top
htop
kill

Packages

apt

Services

systemctl
journalctl

Networking

ip
ping
curl
ss

Remote servers

ssh
scp

Development

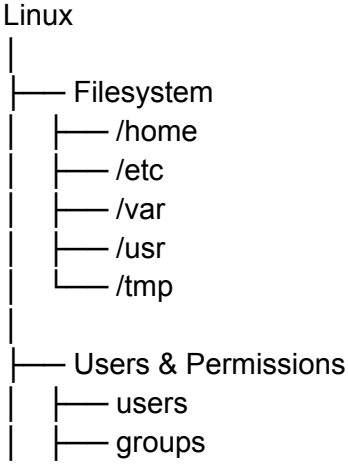
git
python3
pip
node
npm
docker

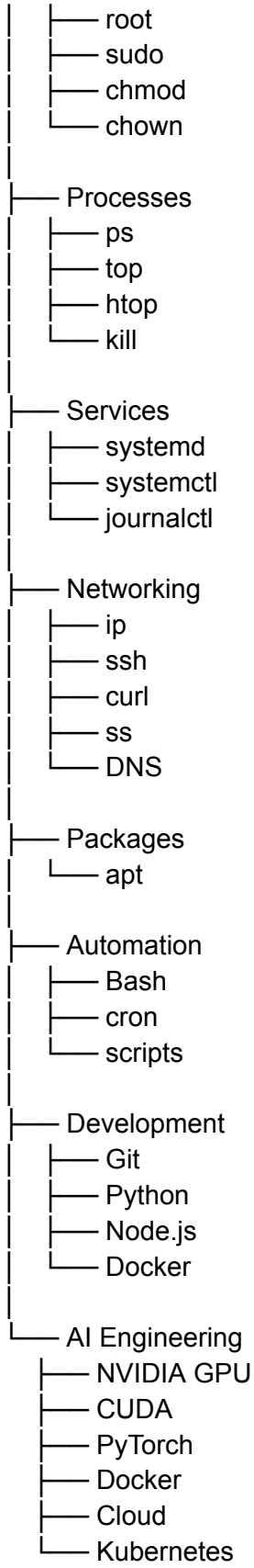
Help

man
--help

---

60. Linux Engineer Mental Model





Priority for an AI/Software Engineer

Learn in this order:

1. pwd / ls / cd
2. mkdir / touch / cp / mv / rm
3. cat / less / head / tail
4. find / grep
5. Absolute & relative paths
6. Users / groups / root
7. sudo
8. chmod / chown
9. Processes / ps / top / kill
10. Environment variables
11. apt
12. systemctl / journalctl
13. Networking / SSH
14. Bash scripting
15. Git
16. Docker
17. Cloud Linux servers
18. Kubernetes
19. NVIDIA / CUDA
20. AI/ML deployment

Core principle: Don't try to memorize every command. Learn the command families and use `man COMMAND` or `COMMAND --help` when you need the exact options.